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Palaeomagnetism, mafic dykes and the Protogine Zone, southern Sweden

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ABSTRACT

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The Protogine Zone (PZ) forms the border between the Svecofennian and Sveconorwegian domains of southern Sweden. Intruded in the Zone are three generations of mafic dykes dated 1510 Ma, 1180 Ma and 930 Ma. A palaeomagnetic study of these dykes in the central and southern parts of the PZ yields two main directions of magnetization: 'A' ($D = 291°$, $I = -78°$, $N = 31$) and 'B' ($D = 129°$, $I = 40°$, $N = 13$). These directions give palaeopoles on the Sveconorwegian Loop of the Fennoscandian apparent polar wander path (APWP). A third, poorly defined group, 'C', yields a mean declinationof 185° and inclination of 9°, with the corresponding palaeopole situated on the c. 1500–1600 Ma part of the APWP. Two other sites yield palaeopoles located in the 1200 Ma part of the path. The data imply that the Sveconorwegian orogeny caused a remagnetization of the two older dyke generations and that this remagnetization was complete in the southern part of the PZ, while a few dykes retained their original magnetization in the central part of the Zone. Either the difference between the 'A' and 'B' directions record appreciable APW movement during differential cooling of the PZ, or the 'A' and 'B' directions may represent an asymmetric reversal of the Earth's magnetic field. In the latter case they may be contemporary and have implications for the size and form of the Sveconorwegian Loop of the Fennoscandian APWP.

Introduction

The general outline of the Fennoscandian apparent polar wander path (APWP) for the Proterozoic is fairly well established (Pesonen et al., 1989). This makes it possible to use it for dating geological events by establishing the characteristic natural remanent magnetization (NRM) in Fennoscandian rock units and correlating the pole positions with the dated APWP. Examples of events that may potentially be dated in this way, are the formation and intrusion of a rock unit, the cooling and uplift of rock units and the regional metamorphism of a terrane. The Sveconorwegian Loop (SL) is an important feature of the Fennoscandian APWP and is mainly based on palaeomagnetic data from rocks affected by the Sveconorwegian orogeny c. 1200–850 Ma ago.

The Protogine Zone (PZ) is a major geological lineament in the southern part of the Fennoscandian Shield (Fig. 1) and palaeomagnetic studies of rock units related to this Zone have given important information on the Sveconorwegian part of the Fennoscandian APWP. New age data on mafic dykes connected with the PZ have been reported (Johansson and Johansson, 1990) which imply the existence of a range of magmatic events in the PZ. In an attempt to further elucidate its history, a palaeomagnetic study has been made on a number of mafic dykes in the central and southern part of the PZ. The results obtained are compared with published palaeomagnetic data in this and adjacent areas (Mulder, 1971; Bylund, 1981, 1985; Stearn and Piper, 1984; Ljungstedt, 1989).

Geology and age relations

The Protogine Zone forms the border between the Svecofennian Transscandinavian Granite–

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Porphyry Belt (TGPB) in the east and the Southwestern Gneiss Complex (SGC) in the west (Gorbatschev, 1980; Gaal and Gorbatschev, 1987; Lindh, 1987; Fig. 1). The TGPB forms the western limit of the Svecofennian domain. Recent U–Pb dating on TGPB granitoids have yielded ages between 1808 to 1769 Ma (Jarl and Johansson, 1988). The SGC probably formed 1760 to

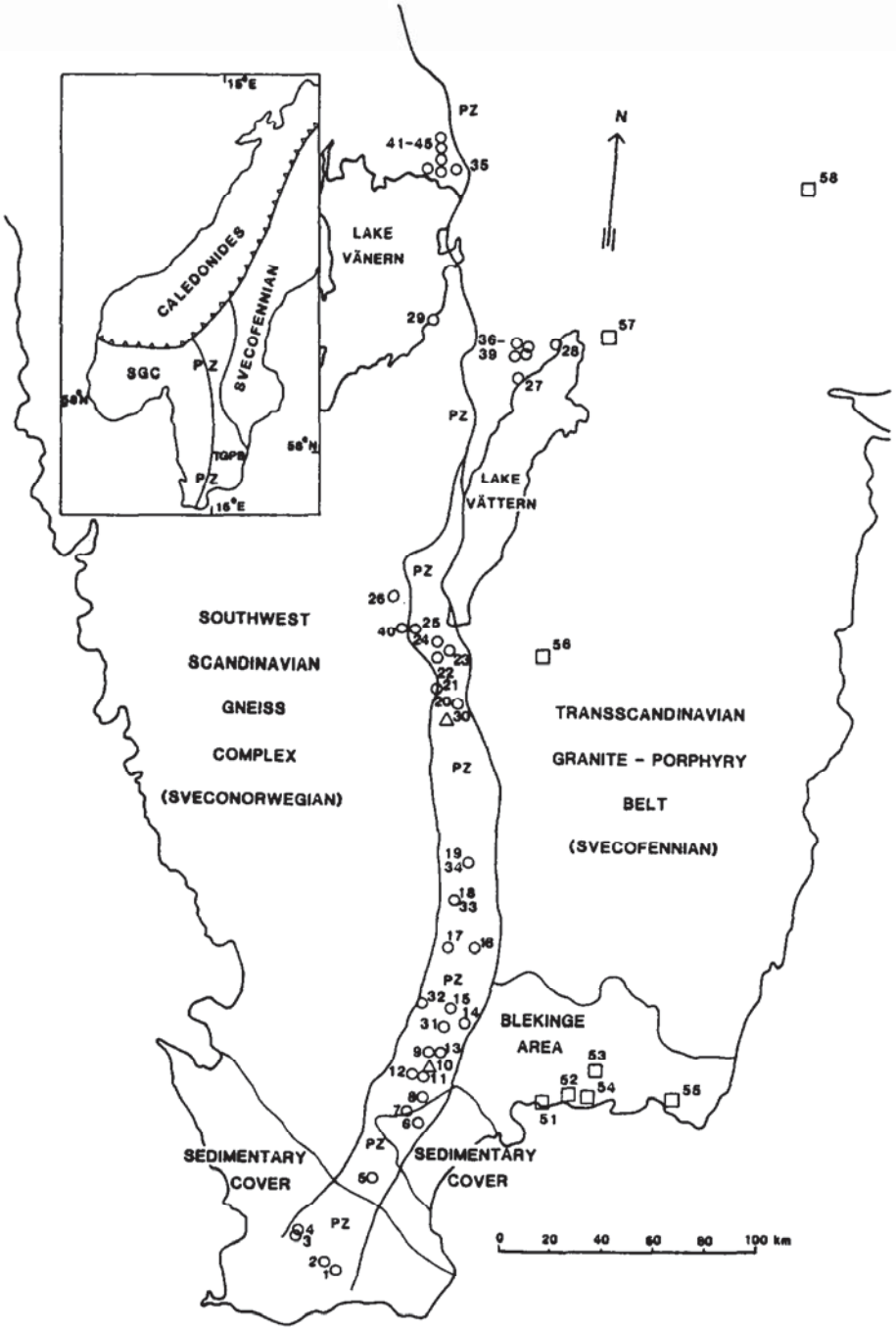


Fig. 1. Geological sketchmap of southern Sweden with palaeomagnetic sample sites. Tectono–magmatic divisions are after Magnusson et al., 1960. Circles denote dolerite and triangles syenite sites in the PZ, squares denote BDD-dolerite sites east of the PZ. The sedimentary cover consists of Phanerozoic rocks.

1600 Ma ago with later additions of both mafic and granitic material. The area is strongly metamorphosed and shows a sequence of deformations which ended with the Sveconorwegian orogeny between c. 1200 and 850 Ma. The Blekinge area in southeastern Sweden (Fig. 1) is probably related to the SGC but escaped the Sveconorwegian metamorphism and deformations that reworked the SGC area.

The TGPB is composed of unmetamorphosed or slightly metamorphosed rocks, while the SGC rocks are metamorphosed into amphibolite to granulite facies (Johansson and Johansson, 1990). The difference in metamorphic grade has been explained by an uplift of the western gneissblock in relation to the eastern granite–porphyry block during Sveconorwegian time, with the already existing PZ acting as a hinge (Welin and Blomqvist, 1966; Patchett and Bylund, 1977).

The PZ, sketched in Fig. 1 (after Magnusson et al., 1960), is a c. 15 to 30 km wide zone of sheared and schistose gneisses of granitic and tonalitic composition. Its borders, however, are not well defined due to lack of outcrops and also because of differing opinions on how to interpret lithologies and tectonic features. In the central and northern parts, the PZ is represented by the border between the TGPB and the SGC, while its real width here probably is comparable with the width outlined in its southern half. The Zone stretches from southernmost Sweden in a N to NNW direction until it disappears under the Caledonian front in Norway (Fig. 1). It has been correlated with the Grenville front of the Laurentian Shield (Patchett and Bylund, 1977; Gower and Owen, 1984) either as a direct continuation, or as the opposite border, of the Grenville–Sveconorwegian terrain. The PZ has been interpreted variously as a collision suture (EUGENO-S, 1988), as a tensional feature, an aborted rift (Andreasson and Rodhe, 1990), or together with the TGPB as a marginal rifted zone of the Svecofennian crust associated with subduction west of the Fennoscandian Shield (Gorbatschev, 1980; Zuber and Öhlander, 1990). Geological similarities across the Zone in its central part and early Sveconorwegian magmatism along its length argue against the interpretation of a Sveconorwegian collision (Gorbatschev, 1980; Johansson, 1991). A discussion of the origin of the PZ is given in Andreasson and Rodhe (1990).

Three age groups of major intrusions related to the PZ have been identified by Johansson and Johansson (1990). These consist of, firstly, the Värmland dolerites dated at c. 1500 Ma, which lie within and west of the central and northern areas of the PZ. Two further age groups dated at c. 1180 and 930 Ma consist of major dykes within the PZ. In addition to these three main groups there are the Blekinge–Dalarna dolerites (BDD) which run parallel and to the east of the PZ. These dolerites are contemporary with or younger than the 930 Ma PZ dolerites (Patchett, 1978; Johansson and Johansson, 1990). The available age data are listed in Table 1.

The PZ dykes are subvertical and range from a few metres to over one hundred metres wide. They include unmetamorphosed dykes with chilled contacts, partly or wholly metamorphosed dykes, and garnet amphibolites. Occasionally, only the margin is amphibolitized leaving a preserved core. Mapping in the southern part of the PZ by the Geological Survey of Sweden (Kornfält et al., 1978; Kornfält and Bergström, 1983; Wikman et al., 1983) presents a selective pattern of tectonics and metamorphism where completely metamorphosed prisms of rocks are surrounded by unaltered rocks. This pattern appears to be typical of the whole PZ. At several sites (e.g. sites 5, 9, 23), the dolerites are cut by dense to fine-grained doleritic dykes generally with the same trend as the main dyke and from a few centimetres to a few metres wide.

Mineralogically there are two varieties of dolerites in the PZ. One set, generally the larger ones, includes olivine, the other does not. The magmatic petrology comprises plagioclase, augite, hypersthene, biotite, apatite and opaques. The degree of secondary alterations varies from perfectly preserved magmatic texture and mineralogy to deformed garnet amphibolite. Descriptions of the petrology, mineralogy, geochemistry and modal compositions are given in Kornfält et al. (1978), Kornfält and Bergström (1983), Wikman et al. (1983), Lindh et al. (1981) and Solyom et al., 1984. Opaques occur mainly as ilmenite and

TABLE 1

Palaeomagnetic results and radiometric ages of intrusions in the Protogine Zone

Site	Rock type	S. lat./S. long.	N	Decl.	Incl.	α_{95}	k	Group	Pol	Radiometric age	Ref.
1	Belinga (D)	55.55/13.62	6	270	−72	10	48	A	N	–	2
2	Stenberget (D)	55.55/13.45	3	198	−87	–	–	A	N	–	2
3	Romeleklint (D)	55.61/13.45	6	250	−82	3	587	A	N	1565 K–Ar	2, 6
4	Krutladan (D)	55.65/13.43	13	286	−84	6	47	A	N	–	2
5	Bjärahall (D)	55.92/13.84	21	347	−84	3	133	A	N	1537, 984 K–Ar	2, 6
6	Balsberget (D)	56.11/14.19	12	146	−89	6	59	A	N	802 K–Ar	2, 6
7	Bjärlöv (D)	56.13/14.10	7	306	−81	6	115	A	N	1179 K–Ar	2, 6
8	Röetved (D)	56.16/14.23	6	107	−87	9	62	A	N	–	2
9a	Rumperöd (D)	56.33/14.13	4	340	−80	6	270	A	N	1470 K–Ar; 1177 Sm–Nd 948 Rb–Sr	2, 6, 8
9b	Rumperöd (D)	56.33/14.13	5	302	−62	3	337	A	N	1177, 1109 K–Ar; 1177 Sm–Nd, 948 Rb–Sr	2, 6
10	Syenite (S)	56.20/13.85	5	303	−82	7	120	A	N	1185, 682 K–Ar; 1184 Rb–Sr	2, 6
11	Broby (D)	56.25/14.07	5	235	−67	12	28	A	N	–	9
12	Olastorp (D)	56.23/14.04	9	328	−79	10	21	A	N	–	9
13	Björkeröd (D)	56.32/14.23	6	232	−68	6	101	A	N	1016, 802 K–Ar; 932 Sm–Nd, 1112 Rb–Sr	6, 8, 9
14	Bagaretorp (D)	56.37/14.44	6	309	−70	6	107	A	N	927 Sm–Nd; 1031 Rb–Sr	8, 9
15	Hökön (D)	56.48/14.30	9	303	−79	4	174	A	N	–	9
16	Peppanäs (D)	56.62/14.38	9	305	−83	7	40	A	N	–	9
17	Taxås Klint (D)	56.63/14.17	6	330	−79	5	120	A	N	–	9
18	Målaskog (D)	56.84/14.25	8	325	−79	5	106	A	N	886 K–Ar	1, 9
19	Hjortsjö (D)	56.94/14.35	3	117	49	8	101	B	R	1573 K–Ar	1, 9
20	Klevaberget (D)	57.53/14.18	6	57	−51	15	15	–	N	–	9
21	Bondstorp (D)	57.58/13.95	8	134	35	9	31	B	R	1177 Sm–Nd	8, 9
22a	Röshult (D)	57.65/14.09	7	341	−83	7	66	A	N	1184 Sm–Nd; 1149 Rb–Sr	8, 9
22b	Röshult (D)	57.65/14.09	7	120	38	21	6	B	R	1184 Sm–Nd; 1149 Rb–Sr	8, 9
22c	Röshult (S)	57.65/14.09	4	102	34	10	53	B	R	–	9
23a	Torsvik (D)	57.67/14.15	8	168	36	11	20	B	R	–	9
23b	Torsvik (G)	57.67/14.15	4	38	−80	9	64	–	N	–	9
23c	Torsvik (G)	57.67/14.15	2	152	27	–	–	B	R	–	9
24a	Sml Tab (D, ore)	57.68/14.07	9	178	−9	14	11	C	R	1177 Sm–Nd	8, 9
24b	Sml Tab	57.68/14.07	4	330	46	13	28	–	N	1177 Sm–Nd	8, 9
24c	Sml Tab	57.68/14.07	6	128	19	12	25	B	R	1177 Sm–Nd	8, 9
25	Husberget (D)	57.73/13.92	3	121	33	–	–	B	R	–	9
26	Vråne (D)	57.84/13.71	8	314	−68	6	60	A	N	–	9
27	Granvik (D)	58.70/14.60	3	105	36	–	–	B	R	–	9
28	Olshammar (D)	58.74/14.75	2	138	43	–	–	B	R	–	9
29	Harnäsudden (D)	58.93/13.96	7	326	−74	8	50	A	N	–	9
30a	Vaggeryd (S)	57.50/14.10	9	131	56	13	13	B	R	1127 Rb–Sr	7, 9
30b	Vaggeryd	57.50/14.10	8	145	−52	17	9	–	R	1127 Rb–Sr	7, 9
31	Hägghult MZI (D)	56.40/14.26	50	327	−27	1	338	A	N	838 K–Ar	1
32	Osby MZK (D)	56.71/14.08	5	337	−53	6	163	A	N	781 K–Ar	1
33	Målaskog MZL (D)	56.84/14.25	5	124	51	2	1892	B	R	886 K–Ar	1
34	Hjortsjö MZM (D)	56.95/14.35	5	322	−72	2	1372	A	N	1573 K–Ar	1
35	K-hamn MZS (D)	59.4/14.2	5	128	39	7	117	B	R	1516 K–Ar	1
36	Unden A (D)	58.7/14.2	5	244	−51	9	79	A	N	–	4
37	Unden B (D)	58.6/14.2	20	263	−51	10	10	A	N	–	4
38	Unden C (D)	58.7/14.6	1	265	−76	–	–	A	N	–	4
39	Unden D (D)	58.6/14.6	2	241	38	–	–	–	R	–	4

TABLE 1 (continued)

Site	Rock type	S. lat./S. long.	N	Decl.	Incl.	α_{95}	k	Group	Pol	Radiometric age	Ref.
40	Bottnaryd H (D)	57.7/13.8	4	247	−55	16	36	A	N	–	4
41	K-hamn 176 (D)	59.4/14.0	6	333	−83	–	39	A	N	–	3
42	K-Hamn 177 (D)	59.4/14.0	6	106	2	–	17	–	R	–	3
43	K-hamn 178 (D)	59.4/14.0	6	328	−77	–	81	A	N	–	3
44	K-hamn 179 (D)	59.4/14.0	5	178	−8	–	22	C	R	–	3
45	K-hamn 180 (D)	59.4/14.0	9	199	−11	–	11	C	R	–	3
Palaeomagnetic results and radiometric ages of the Blekinge–Dalarna dolerites presented by Patchett and Bylund (1977) and Johansson and Johansson (1990) and discussed in the text											
51a	Karlshamn (D)	56.15/14.86	9	130	81	5	116	–	R	936, 871 Rb–Sr	10, 7
51b	Karlshamn (D)	56.15/14.86	8	148	81	2	1182	–	R	936, 871 Rb–Sr	5, 7
52	Tärnö dyke (D)	56.27/15.06	5	313	−42	5	255	–	N	880, 871 Rb–Sr	10, 7
53	Bräkne-Hoby (D)	56.26/15.18	7	309	4	6	99	–	N	975, 880, 844 Rb–Sr 844 Sm–Nd	7, 8, 10
54a	Väby (D)	56.17/15.22	3	123	27	–	–	–	R	–	10
54b	Väby (D)	56.17/15.22	4	137	33	10	79	–	R	–	5
55	Fäjö (D)	56.17/15.69	6	313	3	5	155	–	N	935, 925 Sm–Nd; 809, 800 Rb–Sr	8, 10
56	Nilstorp (D)	57.64/14.83	6	315	−27	10	49	–	N	984 Rb–Sr	10
57	Mårsätter (D)	58.81/15.05	5	94	59	14	30	–	R	–	11
58	Årby (D)	59.27/16.46	6	315	−27	10	49	–	N	995 Rb–Sr	10
Mean group directions											
A	–	–	31	291	−78	4	33	–	N	–	9
B	–	–	13	129	40	8	22	–	R	–	9
A + B combined with B reversed		–	44	303	−51	6	12	–	–	–	9
BDD-dolerites		–	7	133	25	14	14	–	–	–	9

Number refers to dykes sampled. N = number of samples used for calculations. S. lat. and S. long. are geographical coordinates for sites. For sites 35 to 45 the geographical data are calculated from author's figures. α_{95} = semi angle of the(1953) precision parameter. N = normal polarity, R = reversed polarity. References: 1 = Mulder, 1971; 2 = Bylund, 1981; 3 = Stearn and Piper, 1984; 4 = Ljungstedt, 1989; 5 = Poorter, 1975; 6 = Klingspor, 1976; 7 = Patchett, 1978; 8 = Johansson and Johansson, 1990; 9 = this paper; 10 = Patchett and Bylund, 1977;11 = Bylund, 1985.

titanomagnetite in amounts ranging from 5 to 15 weight percent (Hjelmqvist, 1934; Kornfält et al., 1978). The grain size of the opaques varies, from 1 mm down to the resolving power of the microscope (c. 0.5 μm). The plagioclase is clouded by a fine dust, probably fine-grained magnetite, which gives the rocks a black to brown appearance. At site 24, samples were collected in a concentration of olivine and up to 30 percent titanomagnetite, previously mined as an iron ore (Hjelmqvist, 1950).

Sampling and measurement methods

Oriented samples were collected either with a small coredrill or as handsamples. A sun compass was used when possible for in-situ orientation. Samples were taken from fresh outcrops in quarries and roadcuts. Measurements were made by 'Digico' or 'Molspin' spinner magnetometers. Conventional alternating field (AF) and thermal demagnetization techniques were used for isolating the characteristic magnetic remanence in the specimens. Samples were subjected to progressive AF-demagnetization in steps up to peak fields of 150 millitesla (mT) and thermal demagnetization up to 700 °C. Magnetic susceptibility was measured with a Bartington susceptibility meter after thermal demagnetization steps to check for alterations during heating.

Fisher (1953) statistics were used in the calculation of mean directions. Characteristic rema-

nence components were calculated by a least-square analysis method (Torsvik, 1986).

Palaeomagnetic results and comparisons

The results are listed in Table 1. Examples of specimen behaviour during demagnetization are presented in Figs. 2 and 3. The polarities are classed as normal (N) and reversed (R) according to Pesonen and Neuvonen (1981) and given in Table 1. Corresponding palaeopoles are listed in Table 2.

AF-demagnetization tests show several types of magnetic behaviour. A smooth decrease in intensity combined with stable directions (Fig. 2b, site 13) was most frequent in the dolerites with initial intensities in the range 1 to 5×10^3 mA/m. Another common behaviour is found in the dolerites with high initial remanence (10^4 to 10^5 mA/m). In these cases there is a rapid decline in intensity with generally only a few percent left after demagnetization to 10 mT peak field. However, the directions remained stable (Fig. 2b, site 23).

The thermal demagnetization shows two common behaviours. In one there is a gradual decrease in intensity indicating a range of blocking temperatures (Fig. 3b, site 11). The most frequent type, however, is a shoulder type demagnetization curve (Fig. 3b, site 21) with rapid decrease be-

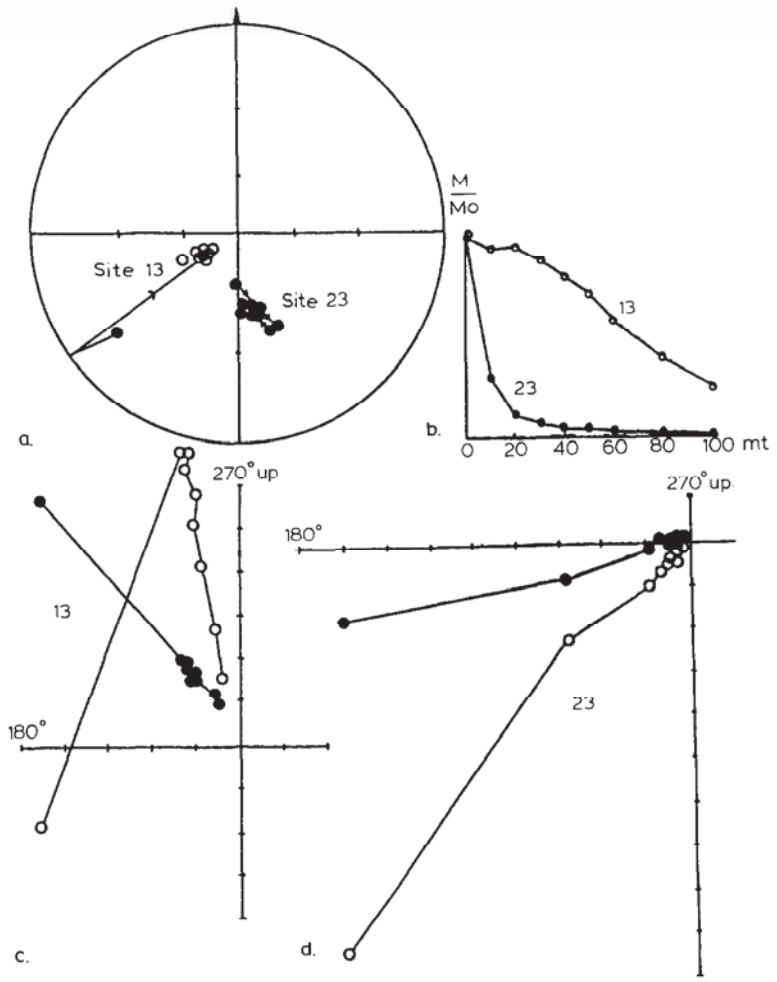


Fig. 2. Examples of behaviour of dolerite specimens during AF- demagnetization: (a) stereoplot, closed (open) symbols denote positive (negative) inclination; (b) normalized curves for the magnetic intensity during demagnetization; (c and d) orthogonal vector diagrams, closed (open) symbols represent vector end-points in the horizontal (vertical) plane.

tween 540 and 570 °C. In both cases, only minor directional changes occur during demagnetization. Above c. 570 °C, the directions become scattered and no consistent high-temperature direction is observed. The susceptibility measurements do not indicate any changes in magnetic mineralogy during stepwise heating.

The demagnetization tests indicate that titanomagnetite with a composition close to pure magnetite is the main carrier of the characteristic remanence as suggested by the microscopic studies, chemical analysis and norm calculations (Kornfält et al., 1978).

It is clear from Fig. 4 that the site mean directions can be divided into groups. These comprise the well-defined groups 'A' and 'B', and a less well defined group 'C' consisting of three directions from sites 24, 44, and 45. Further, two site directions, 20 and 39, are antiparallel and may constitute a fourth group and the same may be the case for the directions 24b and 30b.

All dolerite sites from south of the line X–Y in Fig. 5 yield distinct 'A' directions (Table 1 and Fig. 5). This includes data from the dolerites in the southernmost part of the PZ studied by Bylund (1981). Further, five syenite massifs in the same part of the PZ yield directions very close to those found in the dolerites from the same area (Bylund, 1981). The data from these five syenites are included in entry 10 of Table 1.

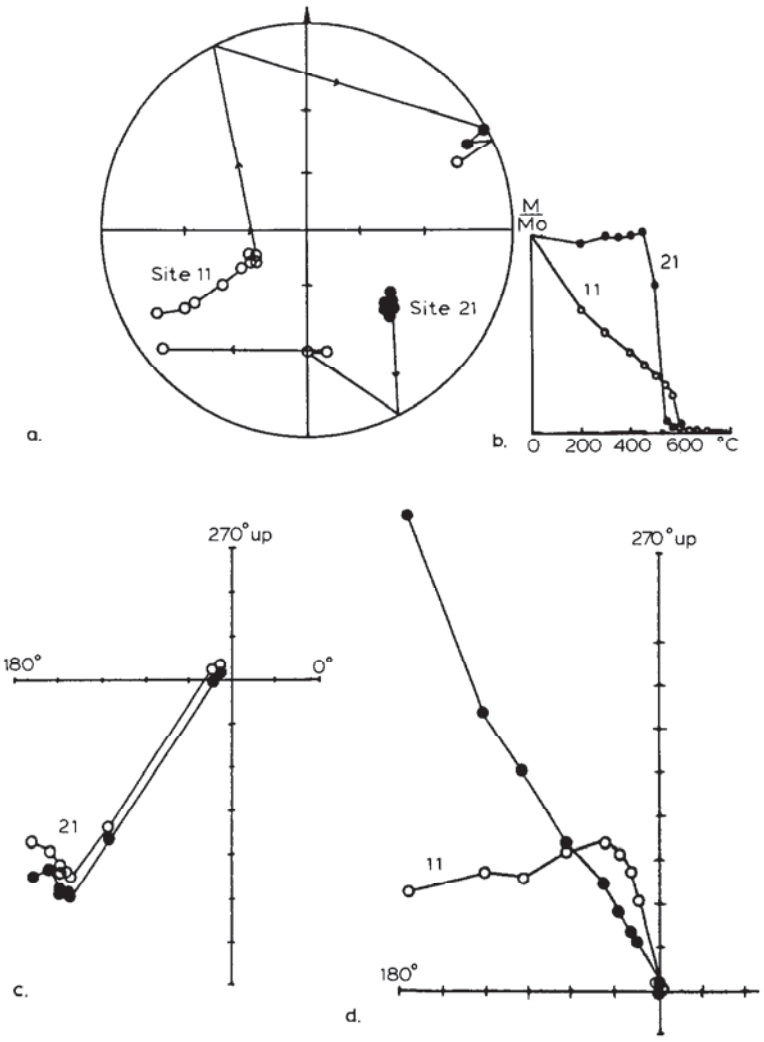


Fig. 3. Examples of behaviour of dolerite specimens during thermal demagnetization: (a) stereoplot; (b) normalized curves for the magnetic intensity during magnetization; (c and d) orthogonal vectordiagram. Symbols as in Fig. 2.

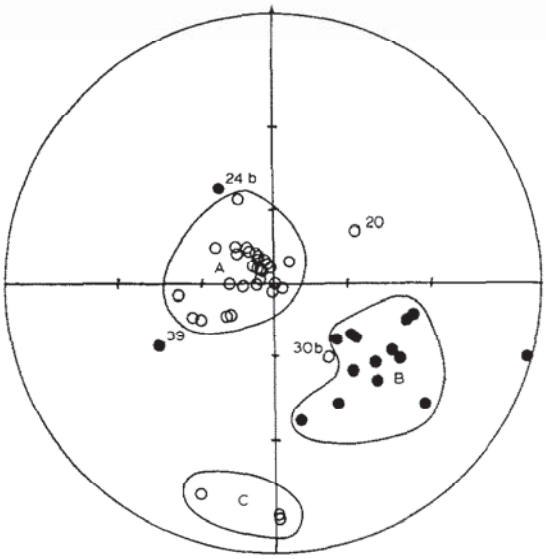


Fig. 4. Stereogram with site mean directions and division into groups recognized after demagnetization. Symbols as in Fig. 2.

A preliminary study of the Vaggeryd syenite (site 30; Bylund in prep.) presents a 'B' direction ($D = 131°$, $I = 56°$) and another direction distinct from the directions recognized here ($D = 145°$, $I = -52°$; Table 1). The Vaggeryd syenite 'B'-direction can be compared with the distinct 'A' directions found in the five sites of similar syenite south of the line X–Y ($D = 302°$, $I = -82°$, entry 10 in Table 1). Site 20 yields a direction $D = 57°$, $I = -51°$; it does not belong to any of the groups recognized elsewhere.

Sites 21 and 23 yields group 'B' directions. At site 23, three samples from the adjacent granite were measured. They were situated between 0.1 to 4 m from the contact with the dolerite. In these, one direction is recognized with steep negative inclination (23b), of group 'A' affinity, and a group 'B' direction (23c) is isolated at high temperatures.

At site 22, both 'A' and 'B' group directions are observed, the latter being less well defined with an α_{95}-value of 15° compared to 6.5° for the 'A'-group data. A positive baked contact test on a fine-grained granitic rock in contact with the dolerite (22c) yields a direction of $D = 102°$, $I = 34°$ close to the B-group directions. The high coercivity and high temperature component indicate that hematite is remanence carrier in this contact rock. In one thermally demagnetized dolerite specimen from this site, a low temperature steep negative 'A' direction was subtracted to define a 'B' direction with intermediate positive inclination (Fig. 6).

At site 24, samples were collected in the titaniferous iron ore that forms a large part of this mafic intrusion (Hjelmqvist, 1950). The iron content is around 30 percent and the ore was mined mainly for the high content of vanadium. The

TABLE 2

Palaeopole positions

Unit	N	P. lat.	P. long.	Palaeolat.	D_p (°)	D_m (°)	Ref.
A	31	45° S	225° E	−68	8	8	9
B	13	1° S	240° E	22	6	10	9
A + B combined	44	27° S	232° E	−51	8	10	9
C	3	35° N	188° E	−5	6	12	9
site 20	6	11° S	148° E	−33	13	20	9
site 39	2	4° S	139° E	−21	–	–	4
site 24b	4	53° N	242° E	27	11	23	9
site 30b	8	56° N	253° E	−33	16	9	9
BDD-poles							
site 51	9	43° S	214° E	−72	9	15	10
BDD-sites without sites 51 and 57	7	10° N	241° E	13	8	10	10, 11

N = number of site means or sample means used for calculation. D_p and D_m are the semi-axes about the pole at the 95% probability level. References as in Table 1.

palaeomagnetic results are complex and three different directions are identified. One was the 'B' direction found as a low coercivity and low temperature component or as the single component in the specimens measured (24c). The second direction (24a) is a 'C'-group direction ($D = 178°$, $I = -9°$), and the third (24b; $D = 330°$, $I = 46°$) does not belong to any recognized group. Directions scattered, α_{95}-values are larger than 10°. It is possible that the last direction is a

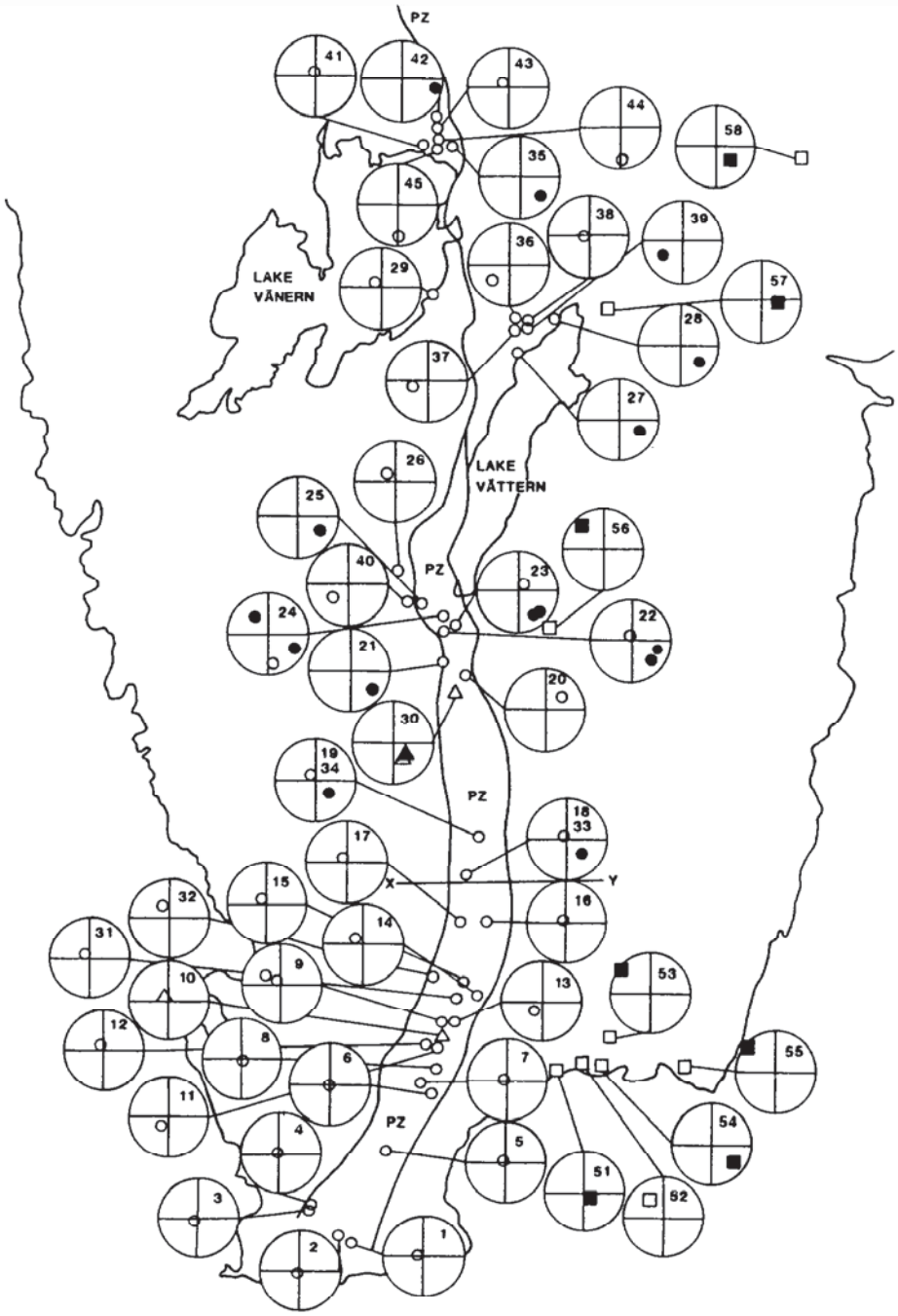


Fig. 5. Site locations and stereograms with site mean directions after demagnetization. The line X–Y denotes the change from only group 'A' magnetization in the south to mixed magnetizations in the central and northern parts of the PZ.

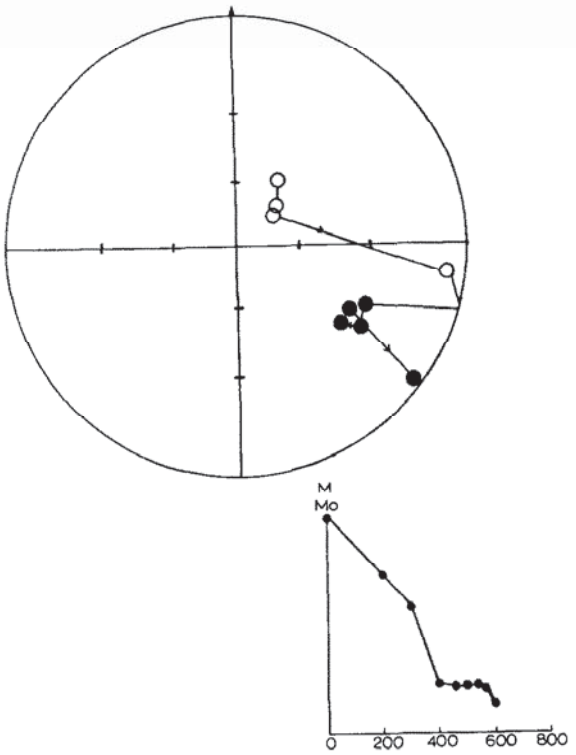


Fig. 6. Thermal demagnetization of a sample from site 22 illustrating the change in direction from 'A' to 'B' with progressive cleaning. Stereoplot (top) and, normalized demagnetization curve (bottom). Symbols as in Fig. 2

reversal to the 'C' directions. Sites 25, 27, and 28 present clear 'B'-group directions, while sites 26 and 29 yield 'A'-group directions.

Ljungstedt, 1989, found an 'A'-group direction in a mafic dyke (site 40) from the same general area as sites 25 and 26. Further, he presented three group 'A' directions in an area west of lake Vättern (Figs. 1 and 5; sites 36–38). Stearn and Piper (1984) made a transect from the Norwegian border eastward to the PZ where they sampled a number of mafic dykes. Of interest here are five sites collected close to or within the PZ (sites 41–45; Fig. 1). Two of these sites, 41 and 43, present group 'A' directions, while the directions of sites 44 and 45 belong to group 'C'. It is not possible at present to assign a group to the shallow direction obtained at site 42. Mulder (1971) obtained a 'B'-group direction in a mafic dyke (site 35) in the same general area as that sampled by Stearn and Piper (1984).

At sites 5, 9 and 23, narrow dykes intruded in the main dyke were sampled. At sites 5 and 23, no significant difference in magnetic directions was found. At site 9, however, where the main dyke is represented by 9a and the intruding dyke by 9b (Table 1), there is a significant difference in directions and this may reflect a difference in age.

The palaeomagnetic results and the APWP; a discussion

The palaeomagnetic data obtained in this study together with previous data from Mulder (1971), Bylund (1981), Stearn and Piper (1984) and Ljungstedt (1989) open possibilities for interpreting the magnetic history of the mafic dykes in the PZ. The radiometric ages provide only limited evidence for the age of the magnetization. The K–Ar ages are scattered and often give data too high due to excess argon (Table 1). The Rb–Sr and Sm–Nd dates give upper possible limits for the magnetization ages. However, it is clear from the dating evidence that there are three generations of mafic dykes connected with the PZ (Johansson and Johansson, 1990). The oldest are the Värmland dolerites west of the PZ in its northern part. They yield ages around 1500 Ma (Welin et al., 1980; Johansson and Johansson, 1990). The dolerites at sites 41–45 and probably also site 35 may belong to this age group (Mulder, 1971; Stearn and Piper, 1984). No age data are available for the dykes west of lake Vättern (Fig. 1). South of the lake, sites 21, 22 and 24 yield Rb–Sr, Sm–Nd and U–Pb ages around 1150 to 1200 Ma. This age interval is broadly coincident with the beginning of the Sveconorwegian orogeny and can be compared with the contemporary or slightly older Central Scandinavian Dolerite Group of central Sweden and western Finland (Gorbatschev et al., 1979; Bylund, 1985). In the southern part of the PZ, Rb–Sr and Sm–Nd dating from sites 9, 13, 14, and 31 yield ages of c. 930 and 1100 Ma, respectively (Johansson and Johansson, 1990). The syenites in the south (entry 10, Table 1) are dated to c. 1200 Ma (Rb–Sr, Klingspor, 1976) and the Vaggeryd syenite (site 30) to 1127 Ma (Rb–Sr; Patchett, 1978).

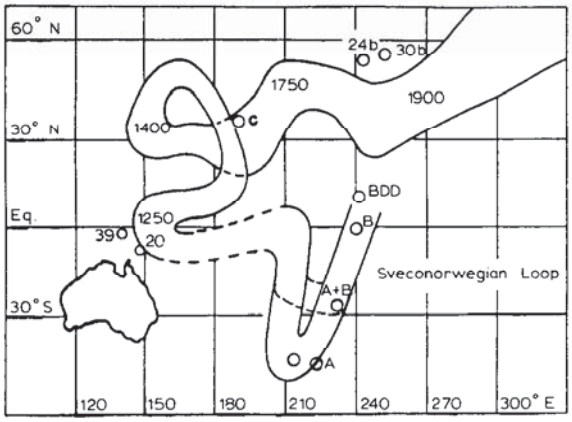


Fig. 7. The Fennoscandian Proterozoic APWP illustrating the open Sveconorwegian Loop (after Pesonen et al., 1989) with palaeopoles from the different PZ mafic dyke groups. A + B is the combined group 'A' and 'B' pole. The dashed lines indicate a possible shortening of the Loop assuming the group 'A' and 'B' directions are of the same age and represent an asymmetric reversal.

Bylund (1981) made a palaeomagnetic study of the mafic dykes at sites 1 to 9 and the five syenite sites combined in entry 10. The result pointed to a complete remagnetization of those rocks in middle to late Sveconorwegian time. The reason for this interpretation was the good agreement in directions and the position of the combined palaeopole at the southern apex of the Sveconorwegian Loop of the Fennoscandian APWP. All these sites yield group 'A' directions. Geological evidence for remagnetization are the presence of garnet and epidote in some of the dykes, e.g. sites 9, 17, 23 and the amphibolitic margins observed at some contacts (Kornfält et al., 1978; Wikman et al., 1983). This demonstrates that post-intrusion metamorphic alteration has taken place. At the same time, some of the dykes appear fresh and are apparently unaffected by later events.

Sites 22 and 39 yield palaeopole positions close to the apex of the c. 1250 Ma loop of the APWP (Fig. 7). This loop is based on palaeomagnetic data from the Central Scandinavian Dolerite Group with an age of c. 1250–1200 Ma (Bylund, 1985). It is possible that the initial NRM in these two dykes has survived the Sveconorwegian remagnetization. These dykes are not dated. The fact that the directions imply a magnetic reversal, reinforces the view that their remanence is primary.

The group 'C' directions give palaeopoles that fall around 1600–1500 Ma on the APWP. Sites 44 and 45 are from the area of the PZ that has yielded SM–Nd and Rb–Sr radiometric ages of c. 1500 Ma while the third direction is one of three directions observed at iron ore site 24. Sites 44 and 45 are the same sites as 179 and 180 of Stearn and Piper (1984). They remark that these rocks have low coercivity and low within-site precision and that the direction may be of Palaeozoic age. However, with regard to the palaeopole position, a middle Proterozoic primary magnetization in accordance with the radiometric dating of the Värmland dolerites cannot be excluded. Johansson and Johansson (1990) present a Sm–Nd isochron for mafic dykes in the PZ including samples from site 24 and this isochron yields an age of c. 1180 Ma. The significance of this age and magnetic direction is obscure at present.

Two anomalous directions observed are the second direction found in the Vaggeryd syenite (site 30b, $D = 145°$, $I = -52°$) and the third direction found in the Smålands Taberg ore (site 24c, $D = 330°$, $I = 46°$). These directions indicate a reversal but the palaeopole falls on the c. 1900 Ma part of the APWP in contradiction with the age determinations. The significance of these magnetic components is uncertain.

The two major groups 'A' and 'B' can be interpreted in three ways: (1) 'A' is older than 'B', (2) 'B' is older than 'A'; or (3) 'A' and 'B' are directions of different polarity but of contemporary age, i.e. they represent an asymmetric reversal of the Earth's magnetic field.

In the APWP of Pesonen et al. (1989; Fig. 7), the group 'A' palaeopoles form the apex of the SL together with the Karlshamn palaeopole of the Blekinge–Dalarna dolerites (site 51, Table 1) and palaeopoles from southern Norway. The 'B'-group palaeopoles are situated in the equatorial part of the SL together with the palaeopoles of the BDD-dolerites. These latter dykes are considered to be the youngest of the dyke generations and constituting the end of the Sveconorvegian orogeny (Patchett and Bylund, 1977; Johansson and Johansson, 1990). This is in accordance with (1) above.

Secondly, if 'B' is older than 'A', then the SL

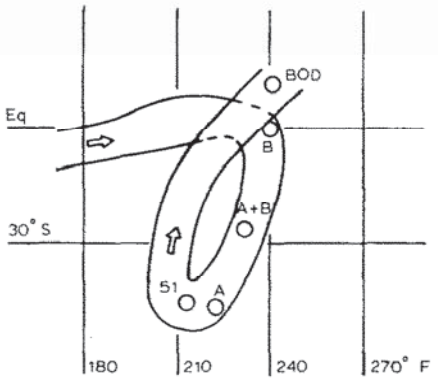


Fig. 8. An alternative interpretation of the Sveconorwegian part of the Fennoscandian APWP with a closed loop (after Bylund, 1981).

is a closed loop with the 'B'-group palaeopoles in the overlapping area as suggested in Fig. 8. Such a loop was suggested by Abrahamsen (1974) and Bylund (1981).

There is a possibility that the groups 'A' and 'B' belong to the same population of magnetic directions but represent both normal and reversed polarities. The mean for group 'A' is $D = 293°$, $I = -78°$ and for 'B' is $D = 129°$, $I = 41°$ (Table 1). This yields a declination difference of 16° and an inclination difference of 37°. Nevanlinna and Pesonen (1983) made a study of asymmetric palaeomagnetic data from the North American Keweenawan igneous activity (1200–1000 Ma). For this period they noted three successive reversals with asymmetric polarities. They explained this in terms of a coaxial two-dipole model comprising a geocentric dipole and a dipole situated at the core–mantle boundary, the latter simulates the spherical harmonic dipole field. Their model suggests that inclinational asymmetries are due to a different nondipole/dipole ratio during the normal and reversed polarities. In this manner, the Laurentian Loop, which is currently mainly based on Keweenawan palaeomagnetic data, has a size which is enhanced by this effect. The 'A' and 'B' directions from the present study may also imply an asymmetric reversal similar to the Keweenawan ones. If they define a reversed field axis, the combined directions of groups 'A' and 'B' give a palaeopole at lat. 27° S and long. 232° E, on the eastern limb of the SL loop (A + B in Table 2 and Figs. 7 and 8). In such a case, the Sveconorwegian Loop may be reduced in real size as sketched in Fig. 7. On the other hand, the late Sveconorwegian BDD-dolerites display symmetric reversals as is evident from entries 52 and 58 in Table 1.

Regional differences and timing of events in the PZ

From Fig. 5 it is evident that sites situated south of the line X–Y all have steep negative inclinations and belong to group 'A'. All rocks studied south of the X–Y line present a single component magnetic pattern, while the magnetic pattern for rocks north of the line is more complicated. Occasionally, both 'A' and 'B'-group directions are found at the same site (e.g. 22 and 23) and in contact rocks. At a few sites, completely different components are recognized (e.g. 20, 39). These results point to a different magnetic history for the rocks in the southern part of the PZ compared with those north of the line X–Y. A simple explanation is that rocks situated south of the line represent a deeper crustal level at the time of intrusion than the rocks north of that line. This would permit the bedrock south of the line to become completely remagnetized during uplift and cooling late in Sveconorwegian time. This model also implies that the bedrock in the PZ north of the line X–Y suffered a more complicated magnetic history and that a selective rate of uplift and cooling of different blocks occurred. This indicates that areas with site directions with old intrusion age are apparently preserved (e.g. site 20) and represent shallow levels of the old crust. It must be emphasized that the line X–Y is approximate and based on magnetic data only. There is at present no tectonic or lithological evidence for a geological change in this area.

In the PZ south of the line X–Y in Fig. 5, mafic dykes and syenites of the 1200 Ma and 950 Ma generations are recognized. The older age groups appear to be all remagnetized and some show signs of post-intrusion metamorphism (e.g. site 9 and site 10). The youngest dyke generation (950 Ma) yield the same 'A'-group magnetic directions as the older ones. This magnetization may be a primary magnetization related to the

intrusion of the dykes but also contemporary with the possible uplift, cooling and remagnetization of this part of the PZ. In this case, the southern part of the PZ would then have acted as a coherent block in contrast to locally restricted movements and intrusions in the central part of the PZ as visualized by the scattered pattern of magnetic directions preserved there.

The data from the youngest dyke generation (950 Ma) in the PZ can be compared with the palaeomagnetic data obtained on the BDD dolerites from east of the PZ (Patchett and Bylund, 1977; Bylund, 1985). Johansson and Johansson (1990) reported Sm–Nd mineral ages from 935 to 844 Ma for two of the dykes included in this study and Rb–Sr mineral ages of 858 to 800 Ma for the same dykes (Table 1). They also reported an age of 840 Ma for one dyke not studied palaeomagnetically. Patchett (1978) reported Rb–Sr mineral and whole-rock ages from 984 to 871 Ma for eight BDD dykes. Hence they are of the same age, or slightly younger, than the youngest PZ dykes. The BDD dykes present directions with shallow to intermediate inclinations of both polarities and palaeopoles in the top part of the SL similar to the 'B'-group data. However, the latter are all of reversed polarity. An exception is the BDD-dyke of Karlshamn, site 51, situated only c. 40 km east of the southern part of the PZ. This dyke has a magnetic direction with steep positive inclination. This magnetic direction can be interpreted as reversed in relation to the 'A'-group direction (Fig. 9). The magnetization in the BDD dykes is considered primary and not acquired during uplift and post-intrusion cooling (Patchett and Bylund, 1977).

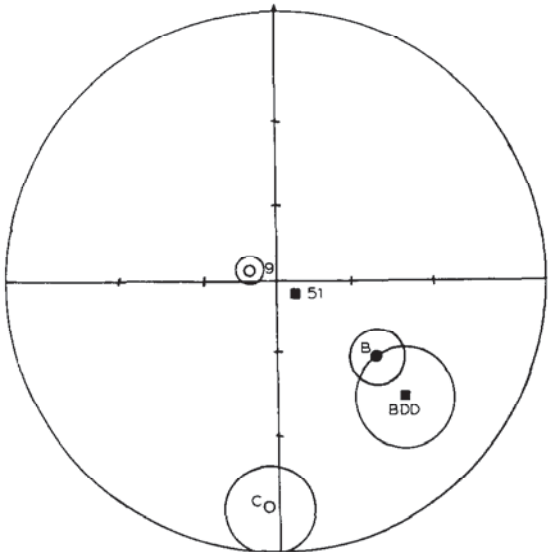


Fig. 9. Stereogram with group mean directions. Note the correspondence between the A-group directions and the reversed direction obtained from the BDD-dolerite site 51. Also note the overlapping confidence circles between the group B mean and the BDD mean. Symbols as in Fig. 2.

There is a difference of about 50° between the 'A'-group mean palaeopole and the corresponding palaeopoles of the 'B'-group and the BDD-dykes. This corresponds to a latitudinal movement of Fennoscandia of c. 5000 km during late Sveconorwegian time.

Conclusions

This study identifies two characteristic NRM directions, 'A' and 'B', of different polarities from mafic dykes within the PZ. The presence of the 'A' direction alone in the southern part of the PZ combined with the radiometric age determinations implies a history of uplift and cooling of the crust in this part of the PZ linked to complete remagnetization. At the same time, another set of dolerites was intruded into the PZ. The uplift and acquisition of the 'A' remanence together with intrusion of the younger dykes was a late Sveconorwegian event at c. 900 Ma ago.

A more complex magnetization history is suggested for the dolerites in the central part of the PZ. Here both 'A' and 'B' directions occur, probably caused by an irregular pattern of uplift and cooling. Furthermore, the 'C' direction and the directions obtained from sites 20 and 39 indicate that the primary NRM in some of the dolerites survived the Sveconorwegian orogeny. This implies intrusion close to surface level with little later uplift of the intruded blocks.

It is plausible that the 'A' and 'B' directions represent a reversal of the ancient field and that they are contemporary in age. Arguing against this is the large asymmetry between the directions, even assuming an asymmetric dipole field, and the regional distribution with only 'A' directions in the southern part of the PZ. However, if

the 'A' and 'B' directions do record a contemporaneous reversal, this will have implications for the size and form of the Sveconorwegian Loop of the Fennoscandian APWP as sketched in Fig. 7. To confirm this, more palaeomagnetic data on successive Sveconorwegian magnetic reversals are needed.

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References

Abrahamsen, N., 1974. The palaeomagnetic age of the WNW-striking dikes around Gothenburg, Sweden. Geol. Fören. Stockholm Förh., 96: 163–170.

Andreasson, P.G. and Rodhe, A., 1990. Geology of the Protogine Zone south of Lake Vättern, southern Sweden: a reinterpretation. Geol. Fören. Stockholm Förh., 112: 127–135.

Bylund, G., 1981. Sveconorwegian palaeomagnetism in hyperite dolerites and syenites from Scania, Sweden. Geol. Fören. Stockholm Förh., 103: 173–182.

Bylund, G., 1985. Palaeomagnetism of middle Proterozoic basic intrusives in central Sweden and the Fennoscandian apparent polar wander path. Precambrian Res., 28: 283–310.

EUGENO-S Working Group, 1988. Crustal structure and tectonic evolution of the transition between the Baltic Shield and the North German Caledonides (the EUGENO-S Project). Tectonophysics, 150: 253–348.

Fisher, R., 1953. Dispersion on a sphere. Proc. R. Soc., London, A 217: 293–305.

Gaal, G. and Gorbatschev, R., 1987. An outline of the Precambrian evolution of the Baltic Shield. Precambrian Res., 35: 15–52.

Gorbatschev, R., 1980. The Precambrian development of southern Sweden. Geol. Fören. Stockholm Förh., 102: 129–136.

Gorbatschev, R., Solyom, Z. and Johansson, I., 1979. The Central Scandinavian Dolerite Group in Jämtland, central Sweden. Geol. Fören. Stockholm Förh., 101: 177–190.

Gower, C.F. and Owen, V., 1984. Pre-Grenvillian and Grenvillian lithotectonic regions in eastern Labrador—correlations with the Sveconorwegian Orogenic Belt in Sweden. Can. J. Earth Sci., 21: 678–693.

Hjelmqvist, S., 1934. Zur Geologie des südschwedischen Grundgebirges, die kristallinen Gesteine des Romeleåses. Medd. Lunds Geol. Mineral. Inst., 58: 1–178.

Hjelmqvist, S., 1950. The titaniferous iron-ore deposit of Taberg in the south of Sweden. Sver. Geol. Unders., C 512: 3–55.

Jarl, L.-G. and Johansson, Å., 1988. U–Pb zircon ages of granitoids from the Småland–Värmland granite–porphyry belt, southern and central Sweden. Geol. Fören. Stockholm Förh., 110: 21–28.

Johansson, Å., 1991. Age of the Önnestad syenite and gneissic granites along the southern part of the Protogine Zone, southern Sweden. In: C.F. Gower, T. Rivers and B. Ryan (Editors), Geol. Assoc. Can. Spec. Pap., 38: 131–148.

Johansson, L. and Johansson, Å., 1990. Isotope geochemistry and age relationships of mafic intrusions along the Protogine zone, southern Sweden. Precambrian Res., 48: 395–414.

Klingspor, I., 1976. Radiometric age determinations of basalts, dolerites, and related syenite in Skåne, southern Sweden. Geol. Fören. Stockholm Förh., 98: 195–216.

Kornfält, K.-A. and Bergström, J., 1983. Beskrivning till berggrundskartan Karlshamn NV (with English summary). Sver. Geol. Unders. Ser. Af, 135, 173 pp.

Kornfält, K.-A., Bergström, J., Carserud, L., Henkel, H. and Sundquist, B., 1978. Beskrivning till berggrundskartan och flygmagnetiska kartan Kristianstad SO (with English summary). Sver. Geol. Unders. Ser. Af, 121, 120 pp.

Lindh, A., 1987. Westward growth of the Baltic Shield. Precambrian Res., 35: 53–70.

Lindh, A., Solyom, Z. and Johansson, I., 1981. The question of chemical homogeneity among basic hypabyssals along the Scandinavian Protogine Zone. Sver. Geol. Unders., C 780: 3–25.

Ljungstedt, G., 1989. Palaeomagnetic age relations of some mafic dykes west of lake Vättern, southern Sweden. Geol. Fören. Stockholm Förh., 111: 21–24.

Magnusson, N.H., Thorslund, P., Brotzen, F., Asklund, B. and Kulling, O., 1960. Description to accompany the map of the pre-Quaternary rocks of Sweden. Sver. Geol. Unders. Ser. Ba, 16: 177 pp.

Mulder, F.G., 1971. Paleomagnetic research in some parts of central and southern Sweden. Sver. Geol. Unders. Ser. C, 653, 56 pp.

Nevanlinna, H. and Pesonen, L.J., 1983. Late Precambrian Keweenawan asymmetric polarities as analyzed by axial offset dipole geomagnetic models. J. Geophys. Res., 88: 645–658.

Patchett, P.J., 1978. Rb/Sr ages of Precambrian dolerites and syenites in southern and central Sweden. Sver. Geol. Unders. Ser. C, 747, 63 pp.

Patchett, P.J. and Bylund, G., 1977. Age of Grenville Belt magnetization: Rb–Sr and palaeomagnetic evidence from Swedish dolerites. Earth Planet. Sci. Lett., 35: 92–104.

Pesonen, L.J. and Neuvonen, K.J., 1981. Palaeomagnetism of the Baltic Shield—implications for Precambrian tectonics. In: A. Kröner (Editor), Precambrian Plate Tectonics. Elsevier, Amsterdam, pp. 623–648.

Pesonen, L.J., Torsvik, T.H., Elming, S.-Å. and Bylund, G., 1989. Crustal evolution of Fennoscandia—palaeomagnetic constraints. Tectonophysics, 162: 27–49.

Poorter, R.P.E., 1975. Palaeomagnetism of Precambrian rocks from southeast Norway and south Sweden. Phys. Earth Planet. Inter., 10: 74–87.

Solyom, Z., Andreasson, P.G., Johansson, I. and Hedvall, R., 1984. Petrochemistry of late Proterozoic rift magmatism in Scandinavia, I. The Blekinge–Dalarna Dolerites (BDD)—volcanism in a failed arm of Iapetus? Lund Publ. Geol., 23: 1–56.

Stearn, J.E.F. and Piper, J.D.A., 1984. Palaeomagnetism of the Sveconorwegian mobile belt of the Fennoscandian Shield. Precambrian Res., 23: 201–246.

Torsvik, T.H., 1986. IAPD—Interactive Analyses of Paleomagnetic Data (User Guide and Program description). Institute of Geophysics, University of Bergen, 74 pp.

Welin, E. and Blomqvist, G., 1966. Further age measurements on radioactive minerals from Sweden. Geol. Fören. Stockholm Förh., 88: 3–18.

Welin, E., Lundegårdh, P.H. and Kähr, A.-M., 1980. The radiometric age of a Proterozoic hyperite diabase in Värmland county, western Sweden. Geol. Fören. Stockholm Förh., 102: 49–52.

Wikman, H., Bergström, J. and Lidmar-Bergström, K., 1983. Beskrivning till berggrundskartan Kristianstad NO (with English summary). Sver. Geol. Unders. Ser. Af, 127, 165 pp.

Zuber, J.A. and Öhlander, B., 1990. Geophysical and geochemical evidence of Proterozoic collision in the western marginal zone of the Baltic Shield. Geol. Rundsch., 79: 1–11.